Module 5: Regularized Regression

Part 1: Ridge Regression

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Regularized Regression (Shrinkage Methods)

- So far, when we fit a model we use the least squares approach.
- Alternatively, we might want to use another fitting procedure instead of least squares.
- These procedures can sometime yield better prediction accuracy and model interpretability.

Least squares estimation

Recall for least squares, we are trying to find:

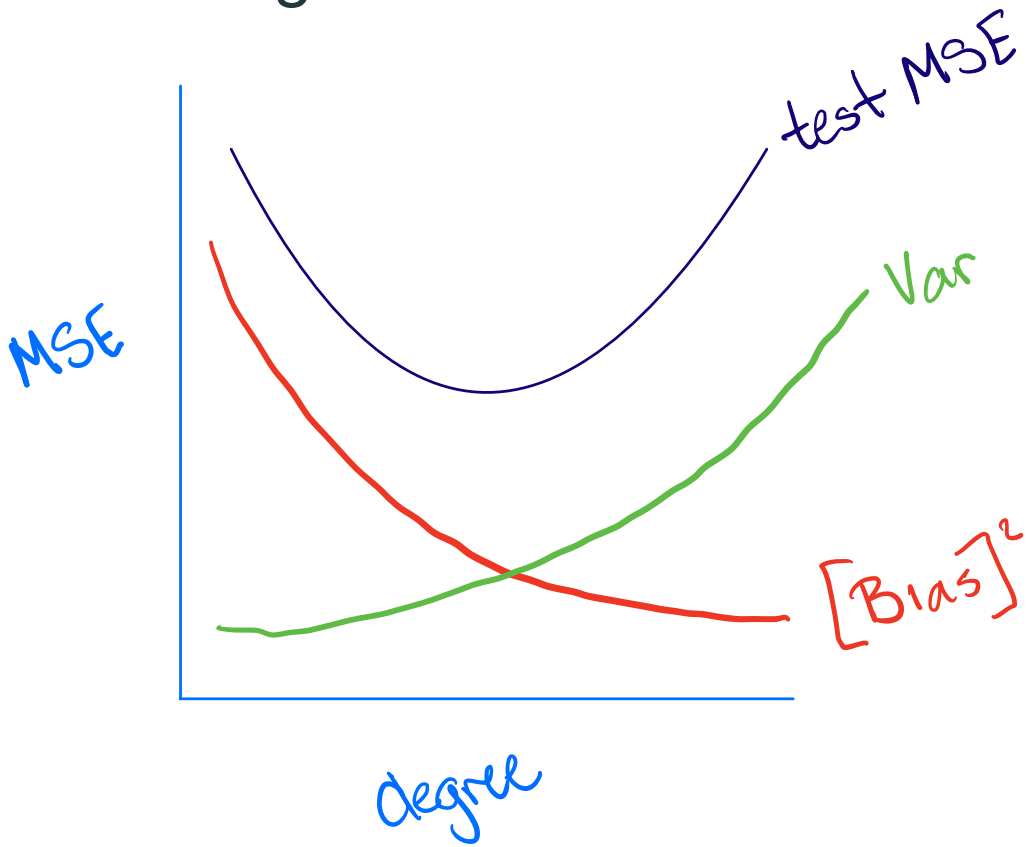
$$\hat{\beta} = \min\left[\sum_{i=1}^{n}\left(y_i - \underbrace{\left[\hat{\beta}_0 + \hat{\beta}_1 \cdot x_{i1} + \hat{\beta}_2 \cdot x_{i2} + \ldots + \hat{\beta}_p \cdot x_{ip}\right]}_{\hat{y}_i}\right)^2\right]$$

Alternative to least squares

- As an alternative, we can fit a model containing all *p* predictors using a technique that **constrains** or **regularizes** the coefficient estimates.
- In effect, this technique will **shrink** some of the coefficient estimates **towards zero**.
- The two best-known techniques for shrinking the regression coefficients towards zero are:
  1. Ridge regression
  2. The lasso

Intuition behind shrinking regression coefficients

We know from the bias/variance decomposition that low bias situations lead to high variance. This in turn, can lead to a high test MSE.



motivation:

↳ introduce a little bias in the model to see if it can reduce the variance

↳ if the reduction in variance is larger than the increase in Bias, we have a lower test MSE

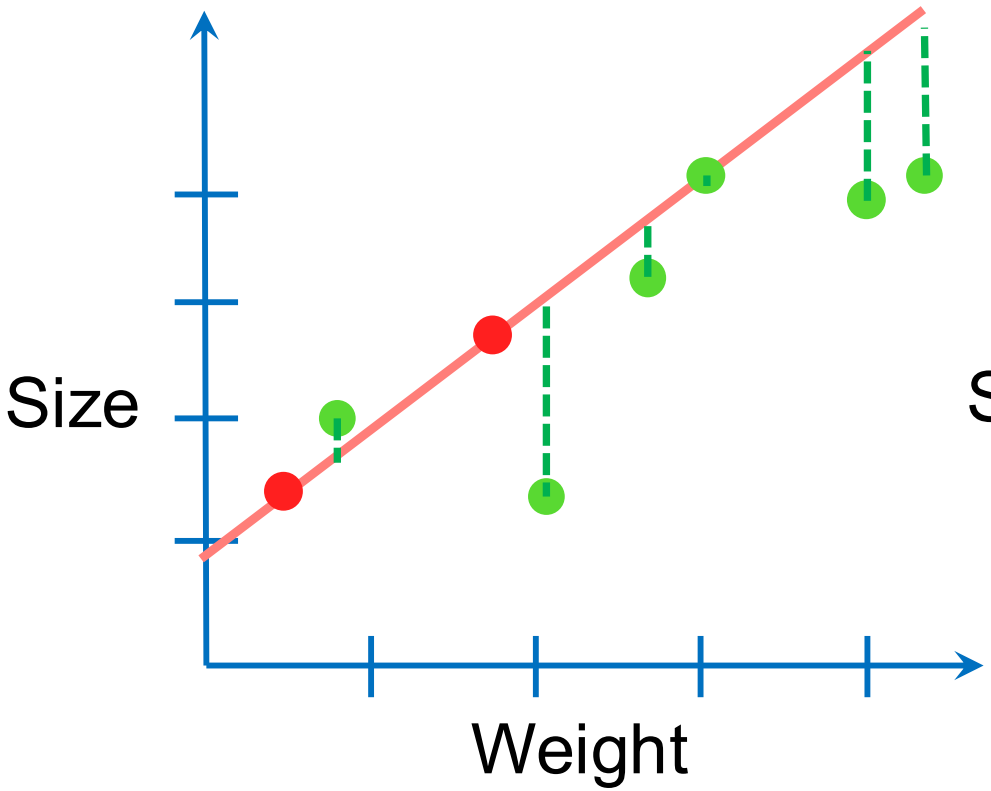
Ridge regression

We want to find $\hat{\beta}^R$ that minimizes

$$\underbrace{\sum_{i=1}^{n}\left[y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \ldots + \hat{\beta}_p x_{ip}\right)\right]}_{\text{RSS}} + \underbrace{\overset{\text{lambda}}{\lambda} \cdot \overbrace{\sum \left(\hat{\beta}_j\right)^2}^{(\text{slope})^2}}_{\text{penalty} \rightarrow \text{source of bias}}$$

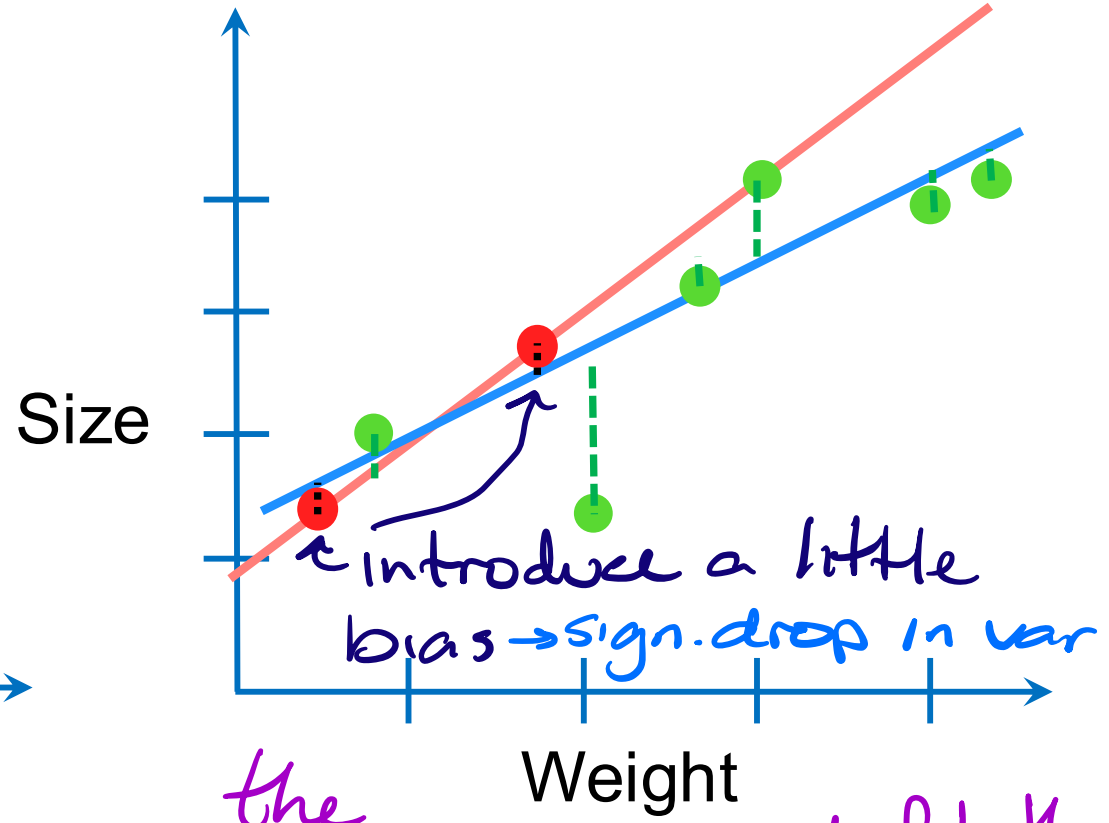
Ridge regression versus least squares

Red dots: training data
Green dots: test data



0 train MSE
large test MSE
→ over fit

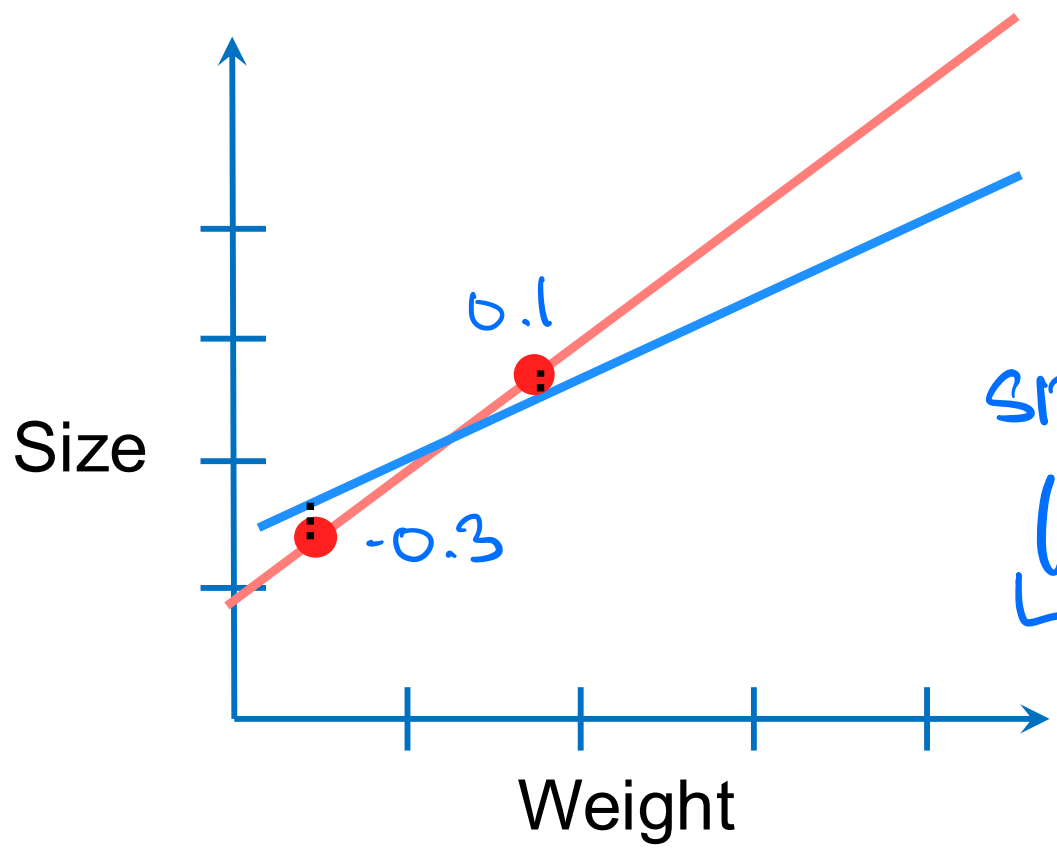
Red line: least squares
Blue line: Ridge regression



the blue line doesn't fit the data as well as the red.

Ridge regression versus least squares

Red line: least squares
Blue line: Ridge regression



Least squares:

$\text{size} = 0.4 + 1.3 \cdot \text{weight}$

$0 + 1 \cdot 1.3^2 = 1.69$

$\text{size} = 0.9 + 0.8 \cdot \text{weight}$

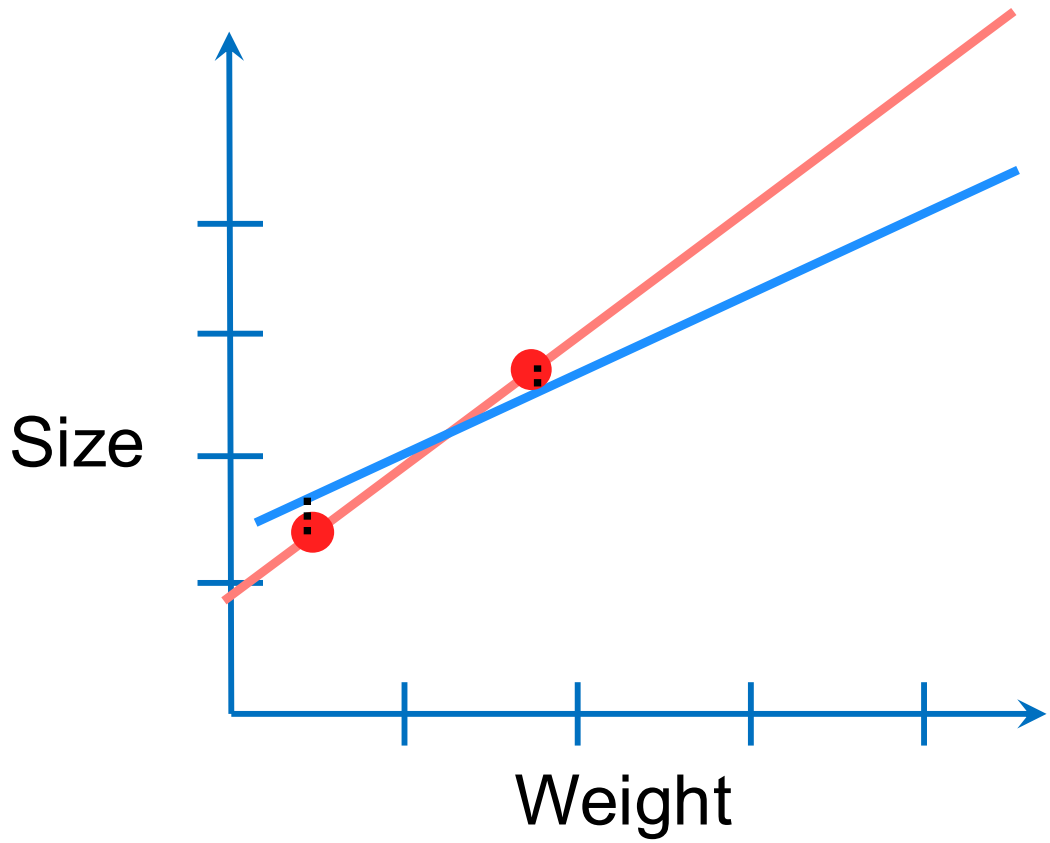
$\underbrace{(-0.3)^2 + (0.1)^2}_{\text{RSS}} + 1 \cdot 0.8^2$

$= 0.74$

Ridge regression versus least squares

Red line: least squares
Blue line: Ridge regression

$$\text{Ridge: RSS} + \lambda * slope^2$$

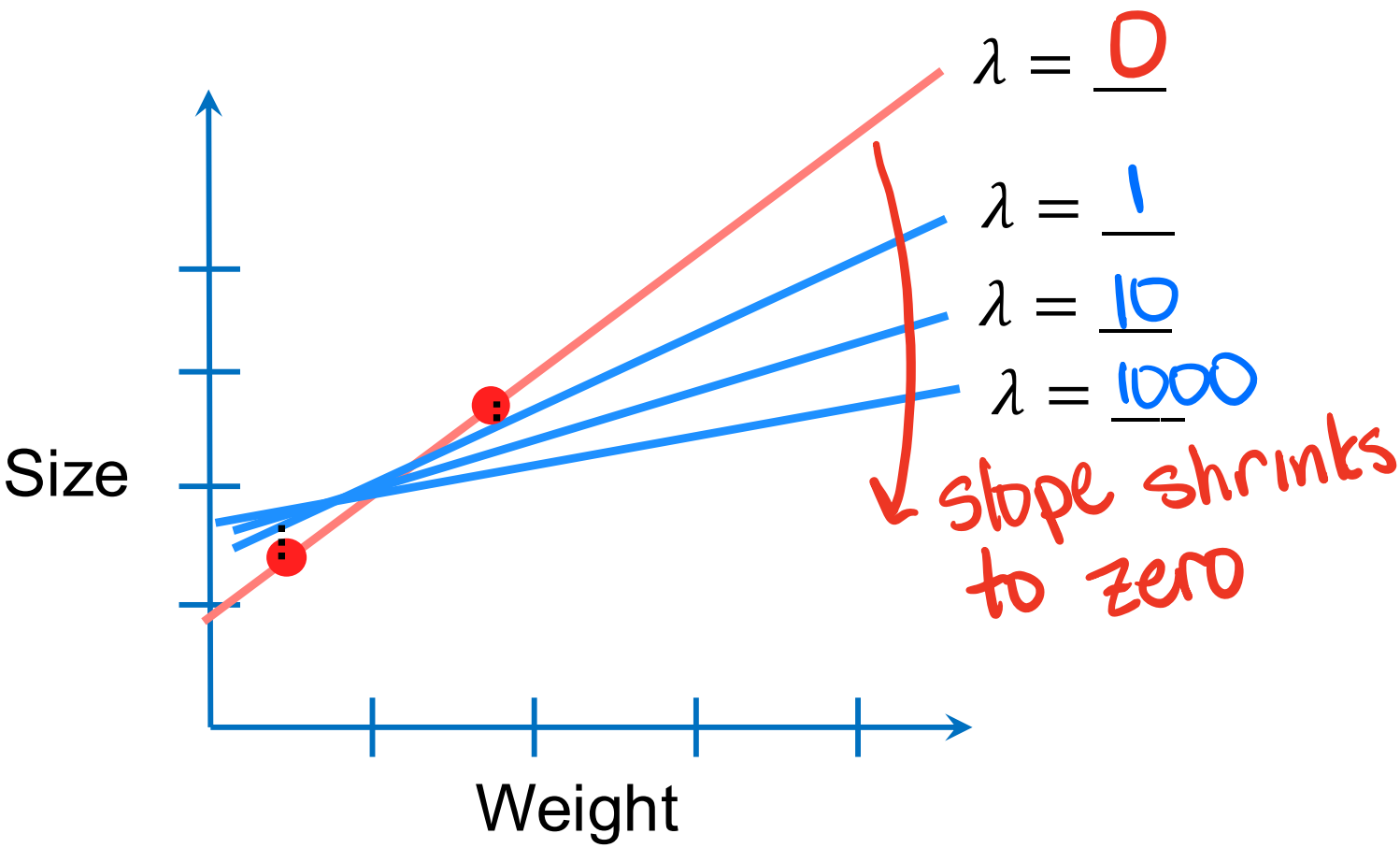


Ridge regression

Red line: least squares
Blue line: Ridge regression

large $\lambda \rightarrow$ smaller coefficients

0, 1,100,1000



Tuning parameter

We want to find $\hat{\beta}^R$ that minimizes

$$\hat{\beta}^R = \min_{\beta}\left(\sum_{i=1}^{n}\left(Y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p\right)\right)^2 + \lambda \sum_{j=1}^{p} \hat{\beta}_j^2\right), \lambda > 0$$

- λ is referred to as our tuning parameter.
- It controls the relative balance of these two terms: modulates the tradeoff between fit and shrinkage.
- $\lambda = 0$: least squares (RSS dominate)
- $\lambda \to \infty$: $\hat{\beta}^R$ (penalty term dominate)

Tuning parameter

- Each value of λ will give you a different set of coefficient estimates.
- Selecting good λ is critical to ridge regression.

λ too small: not reduce variance enough

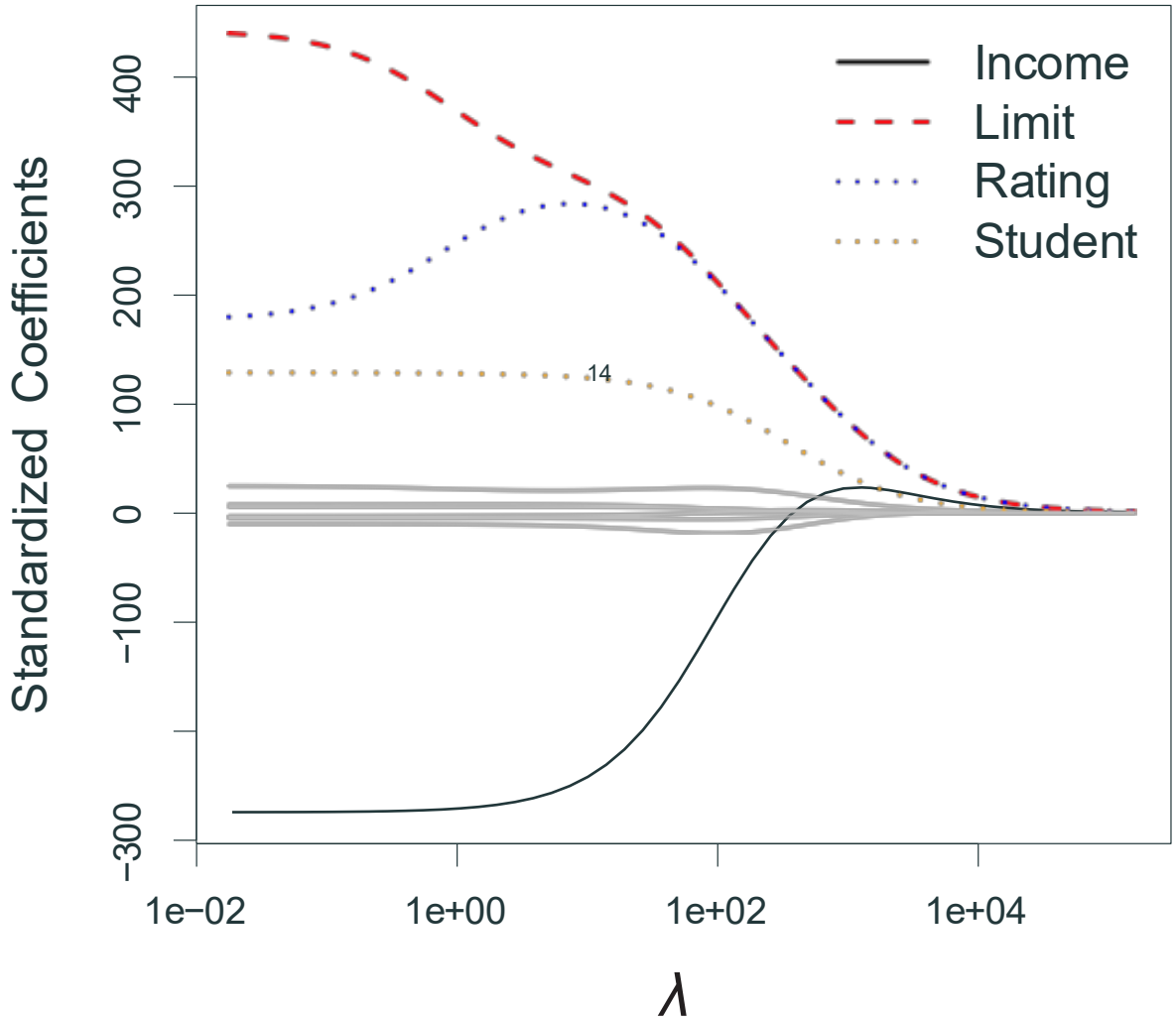
λ too large: over shrink (lose information)

Some things to keep in mind

- The penalty term is applied to the regression coefficients ($\beta_1, \beta_2, \beta_3, \ldots, \beta_p$) but not the intercept.
  - We want to shrink the estimated association of each predictor with the response; however, we do not want to shrink the intercept.
- Before applying ridge regression, we must **standardize the predictors** so they are all on the same scale.
- As we increase λ, does the model become more flexible or less flexible?

How to choose λ ?

Each λ will give us a different set of regression coefficients.



How to choose λ?

split data into training + test

Use cross-validation:

training

1. Choose a grid of λ values.
2. Compute the cross-validation error for each value of λ.
3. Select the value of λ for which the cross-validation error is the smallest.
4. Refit the model using all data and the selected λ. This is your final model.

training + test

See R script: `shrinkage_methods.R`